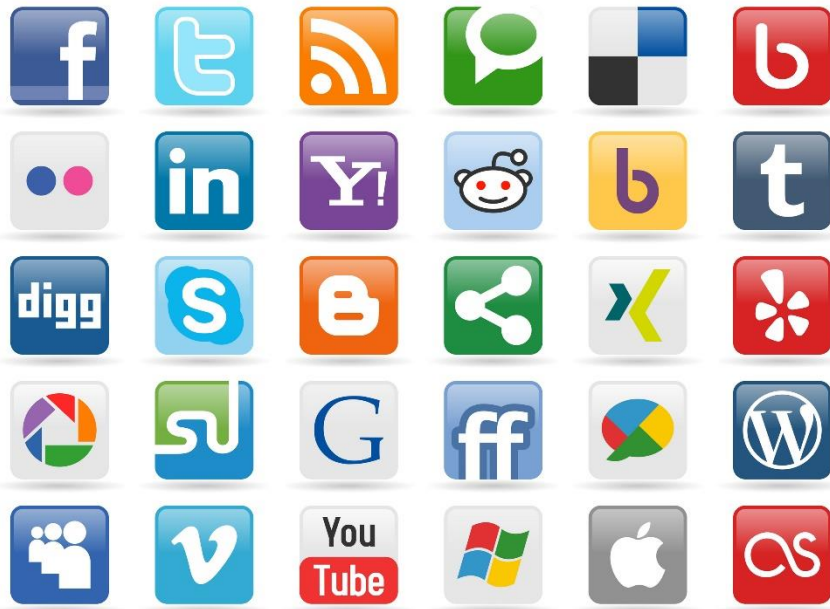




**Network
Measures**

SOCIAL

MEDIA

MINING



Why Do We Need Measures?

- Who are the central figures (influential individuals) in the network?
 - **Centrality**
- What interaction patterns are common in friends?
 - **Reciprocity** and **Transitivity**
- Who are the like-minded users and how can we find these similar individuals?
 - **Similarity**
- To answer these and similar questions, one first needs to define measures for quantifying **centrality**, **level of interactions**, and **similarity**, among others.

Centrality

Centrality in terms of those who you are connected to

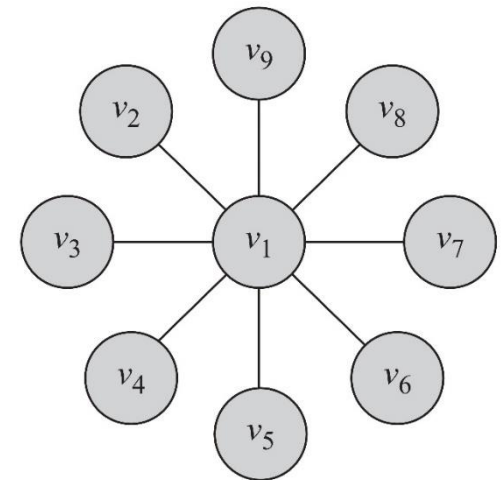
Degree Centrality

- **Degree centrality:** ranks nodes with more connections higher in terms of centrality

$$C_d(v_i) = d_i$$

- d_i is the degree (number of friends) for node v_i

In this graph, degree centrality for node v_1 is $d_1=8$ and for all others is $d_j = 1, j \neq 1$



Degree Centrality in Directed Graphs

- In directed graphs, we can either use the in-degree, the out-degree, or the combination as the degree centrality value:
- In practice, mostly in-degree is used.

$$C_d(v_i) = d_i^{\text{in}} \quad (\textit{prestige})$$

$$C_d(v_i) = d_i^{\text{out}} \quad (\textit{gregariousness})$$

$$C_d(v_i) = d_i^{\text{in}} + d_i^{\text{out}}$$

d_i^{out} is the number of outgoing links for node v_i

Normalized Degree Centrality

- Normalized by the maximum possible degree

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

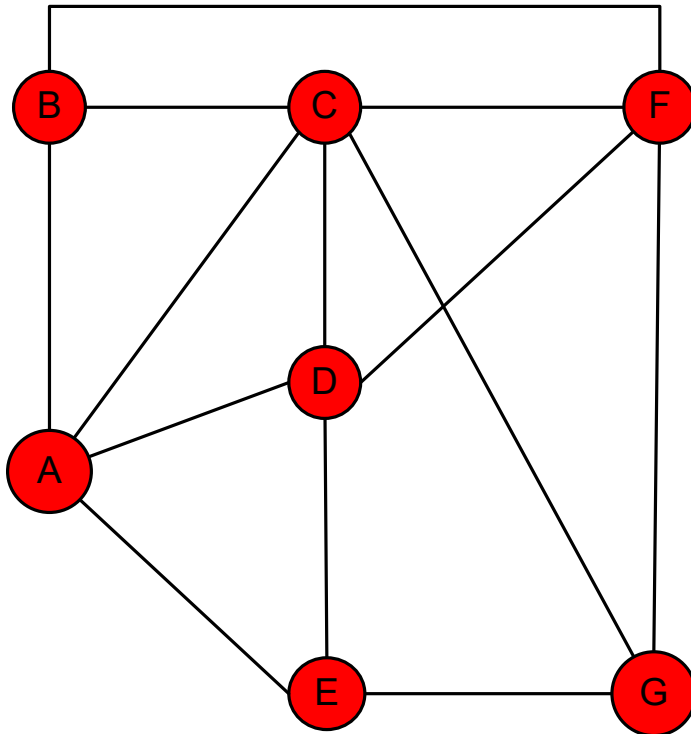
- Normalized by the maximum degree

$$C_d^{\text{max}}(v_i) = \frac{d_i}{\max_j d_j}$$

- Normalized by the degree sum

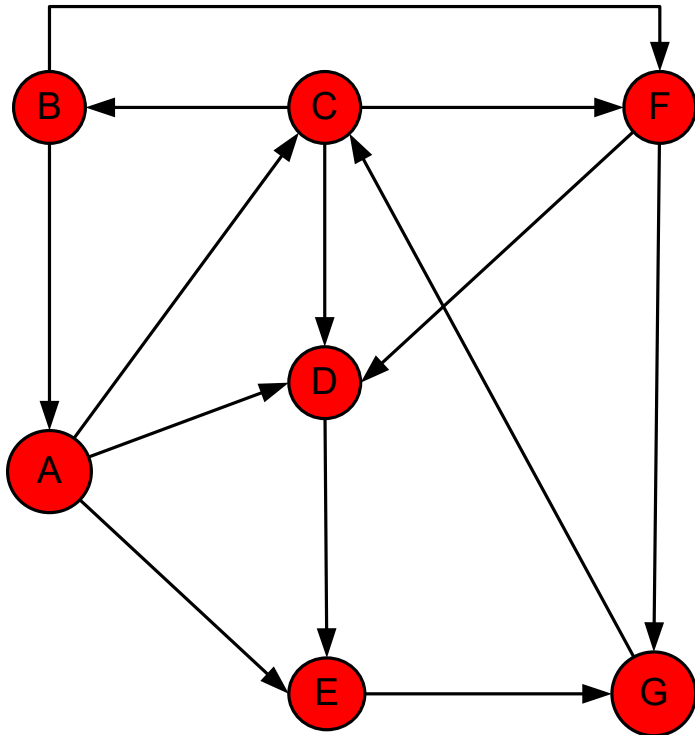
$$C_d^{\text{sum}}(v_i) = \frac{d_i}{\sum_j d_j} = \frac{d_i}{2|E|} = \frac{d_i}{2m}$$

Degree Centrality (undirected Graph) Example



Node	Degree	Centrality	Rank
A	4	$2/3$	2
B	3	$1/2$	5
C	5	$5/6$	1
D	4	$2/3$	2
E	3	$1/2$	5
F	4	$2/3$	2
G	3	$1/2$	5

Degree Centrality (Directed Graph)Example



Node	In-Degree	Out-Degree	Centrality	Rank
A	1	3	1/2	1
B	1	2	1/3	3
C	2	3	1/2	1
D	3	1	1/6	5
E	2	1	1/6	5
F	2	2	1/3	3
G	2	1	1/6	5

Normalized by the maximum possible degree

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

Eigenvector Centrality

- Having more friends does not by itself guarantee that someone is more important
 - Having more **important friends** provides a stronger signal
- Eigenvector centrality generalizes degree centrality by incorporating the importance of the neighbors (undirected)
- For directed graphs, we can use incoming or outgoing edges



Phillip Bonacich

Formulation

- Let's assume the eigenvector centrality of a node is $c_e(v_i)$ (**unknown**)
- We would like $c_e(v_i)$ to be higher when **important** neighbors (**node v_j with higher $c_e(v_j)$**) point to us
 - Incoming or outgoing neighbors?
 - For incoming neighbors $A_{j,i} = 1$
- We can assume that v_i 's centrality is proportional to the summation of its neighbors' centralities

$$c_e(v_i) = \frac{1}{\lambda} \sum_{j=1}^n A_{j,i} c_e(v_j)$$

λ : some fixed constant

Eigenvector Centrality (Matrix Formulation)

- Let $\mathbf{C}_e = (C_e(v_1), C_e(v_2), \dots, C_e(v_n))^T$

$$\rightarrow \lambda \mathbf{C}_e = \mathbf{A}^T \mathbf{C}_e$$

- This means that $\mathbf{C}_e$ is an eigenvector of adjacency matrix $\mathbf{A}^T$ (or $\mathbf{A}$ when undirected) and λ is the corresponding eigenvalue
- Which eigenvalue-eigenvector pair should we choose?

Finding the eigenvalue by finding a fixed point...

- Start from an initial guess $C_e(0)$ (e.g., all centralities are 1) and iterative t times

$$C_e(t) = (A^T)^t C_e(0)$$

- We can write $C_e(0)$ as a linear combination of eigenvectors v_i 's of the A^T

$$C_e(0) = \sum_i \alpha_i v_i$$

- Substituting this, we get

$$C_e(t) = (A^T)^t \sum_i \alpha_i v_i = \sum_i \alpha_i \lambda_i^t v_i = \lambda_1^t \sum_i \alpha_i \left(\frac{\lambda_i}{\lambda_1}\right)^t v_i$$

λ_1 is the largest eigenvalue

Finding the eigenvalue by finding a fixed point...

- As t grows, we will have in the limit

$$C_e(t) \rightarrow \alpha_1 \lambda_1^t v_1$$

- The limiting vector is proportional to the leading eigenvector
- Or equivalently, the centrality $A^T C_e = \lambda_1 C_e$
- If we start with an all positive $C_e(0)$ all $C_e(t)$'s will be positive (why?)
 - All the centrality values would be positive
 - We need an eigenvalue-eigenvector pair that guarantees all centralities have the same sign
 - E.g., for comparison purposes

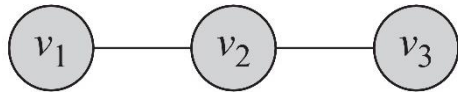
Eigenvector Centrality,.

Theorem 1 (Perron-Frobenius Theorem). *Let $A \in \mathbb{R}^{n \times n}$ represent the adjacency matrix for a [strongly] connected graph or $A : A_{i,j} > 0$ (i.e. a positive n by n matrix). There exists a positive real number (Perron-Frobenius eigenvalue) $\lambda_{\max}$, such that $\lambda_{\max}$ is an eigenvalue of A and any other eigenvalue of A is strictly smaller than $\lambda_{\max}$. Furthermore, there exists a corresponding eigenvector $\mathbf{v} = (v_1, v_2, \dots, v_n)$ of A with eigenvalue $\lambda_{\max}$ such that $\forall v_i > 0$.*

So, to compute eigenvector centrality of A ,

1. We compute the eigenvalues of A
2. Select the largest eigenvalue λ
3. The corresponding eigenvector of λ is $\mathbf{C}_e$.
4. Based on the Perron-Frobenius theorem, all the components of $\mathbf{C}_e$ will be positive
5. The components of $\mathbf{C}_e$ are the eigenvector centralities for the graph.
6. Most software use computation based on power method

Eigenvector Centrality: Example 1



$$\lambda \mathbf{C}_e = A \mathbf{C}_e \quad (A - \lambda I) \mathbf{C}_e = 0 \quad \mathbf{C}_e = [u_1 \ u_2 \ u_3]^T$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 - \lambda & 1 & 0 \\ 1 & 0 - \lambda & 1 \\ 0 & 1 & 0 - \lambda \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 0 - \lambda & 1 & 0 \\ 1 & 0 - \lambda & 1 \\ 0 & 1 & 0 - \lambda \end{vmatrix} = 0$$

$$(-\lambda)(\lambda^2 - 1) - 1(-\lambda) = 2\lambda - \lambda^3 = \lambda(2 - \lambda^2) = 0$$

Eigenvalues are

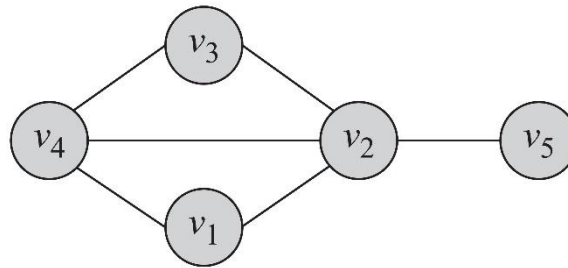
$$(-\sqrt{2}, 0, +\sqrt{2})$$

Corresponding eigenvector (assuming $\mathbf{C}_e$ has norm 1)

Largest Eigenvalue

$$\begin{bmatrix} 0 - \sqrt{2} & 1 & 0 \\ 1 & 0 - \sqrt{2} & 1 \\ 0 & 1 & 0 - \sqrt{2} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{C}_e = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ \sqrt{2}/2 \\ 1/2 \end{bmatrix}$$

Eigenvector Centrality: Example 2



$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightarrow \lambda = (2.68, -1.74, -1.27, 0.33, 0.00)$$

$\nwarrow$ Eigenvalues Vector

$$\lambda_{\max} = 2.68 \rightarrow C_e = \begin{bmatrix} 0.4119 \\ 0.5825 \\ 0.4119 \\ 0.5237 \\ 0.2169 \end{bmatrix}$$

V₁
V₂
V₃
V₄
V₅

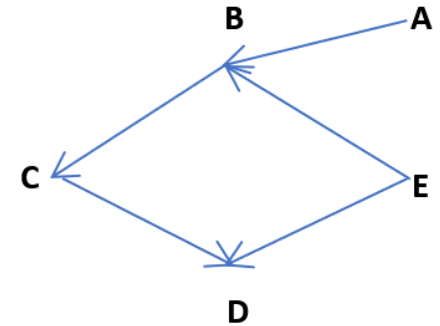
Katz Centrality

- A major problem with eigenvector centrality arises when it deals with directed graphs
- Centrality only passes over *outgoing* edges
- In special cases such as when a node is in a directed acyclic graph centrality becomes zero
 - The node with zero centrality can have many edges connected to it



Elihu Katz

- In the adjacent figure
- $c_A, c_E = 0$, no incoming link
- $c_B = 0$, incoming links from c_A and c_E
- $c_C = 0$, only incoming link from c_B
- $c_D = 0$ incoming links from c_C and c_E



- To resolve this problem, we add bias term β to the centrality values for all nodes
- Giving free centrality value to each node

Eigenvector Centrality

$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^n A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

Katz Centrality, cont.

$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^n A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

Controlling term Bias term

Rewriting equation in a vector form

$$\mathbf{C}_{\text{Katz}} = \alpha A^T \mathbf{C}_{\text{Katz}} + \beta \mathbf{1}$$

vector of all 1's

Katz centrality: $\mathbf{C}_{\text{Katz}} = \beta (\mathbf{I} - \alpha A^T)^{-1} \cdot \mathbf{1}$

Katz Centrality, cont.

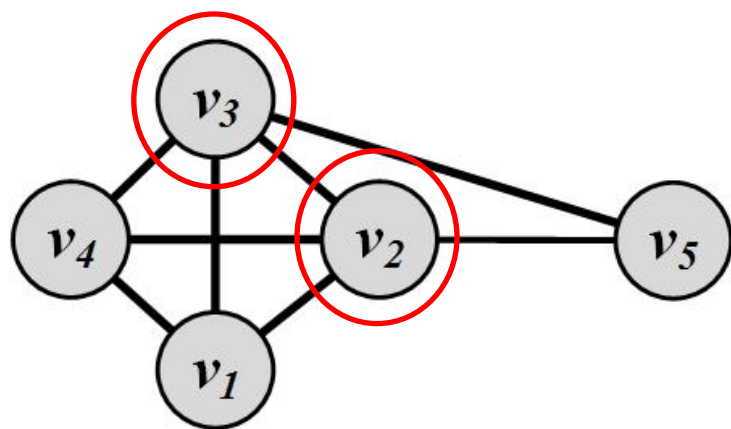
$$C_{\text{Katz}}(v_i) = \alpha \sum_{j=1}^n A_{j,i} C_{\text{Katz}}(v_j) + \beta$$

- When $\alpha=0$, the eigenvector centrality is removed, and all nodes get the same centrality value β
 - As α gets larger the effect of β is reduced

In practice we select $\alpha < 1/\lambda$, where λ is the largest eigenvalue of A^T

Power method vs matrix inversion

Katz Centrality Example



$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} = A^T$$

- The Eigenvalues are: -1.68, -1.0, -1.0, 0.35, **3.32**
- We assume $\alpha = 0.25 < \frac{1}{3.32}$ and $\beta = 0.2$

$$C_{Katz} = \beta(\mathbf{I} - \alpha A^T)^{-1} \cdot \mathbf{1} = \begin{bmatrix} 1.14 \\ \mathbf{1.31} \\ \mathbf{1.31} \\ 1.14 \\ 0.85 \end{bmatrix}$$

**Most
important
nodes!**

- Problem with Katz Centrality:
 - In directed graphs, once a node becomes an authority (high centrality), it passes **all** its centrality along **all** of its out-links
- This is less desirable since not everyone known by a well-known person is well-known
- **Solution?**
 - We can divide the value of passed centrality by the number of outgoing links, i.e., out-degree of that node
 - Each connected neighbor gets a fraction of the source node's centrality

PageRank, cont.

$$C_p(v_i) = \alpha \sum_{j=1}^n A_{j,i} \frac{C_p(v_j)}{d_j^{\text{out}}} + \beta$$

What if the
degree is
zero?



$$\begin{cases} d_j^{\text{out}} > 0 \\ D = \text{diag}(d_1^{\text{out}}, d_2^{\text{out}}, \dots, d_n^{\text{out}}) \end{cases} \rightarrow \mathbf{C}_p = \alpha A^T D^{-1} \mathbf{C}_p + \beta \mathbf{1}$$

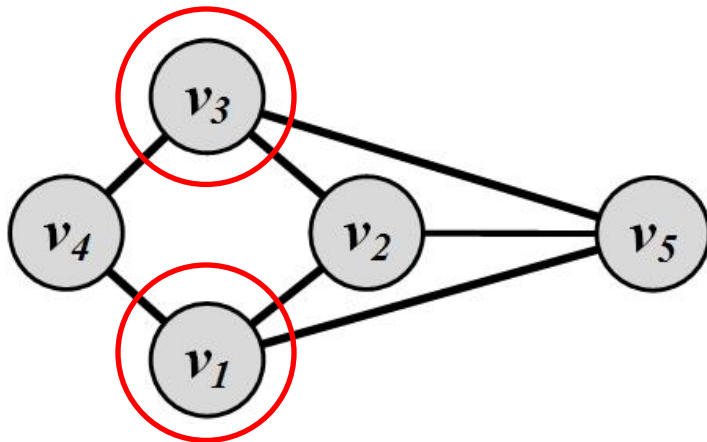


$$\mathbf{C}_p = \beta (\mathbf{I} - \alpha A^T D^{-1})^{-1} \cdot \mathbf{1}$$

Similar to Katz Centrality, in practice, $\alpha < 1/\lambda$, where λ is the largest eigenvalue of $A^T D^{-1}$. In undirected graphs, the largest eigenvalue of $A^T D^{-1}$ is $\lambda = 1$; therefore, $\alpha < 1$.

PageRank Example

- We assume $\alpha = 0.95 < 1$ and $\beta = 0.1$

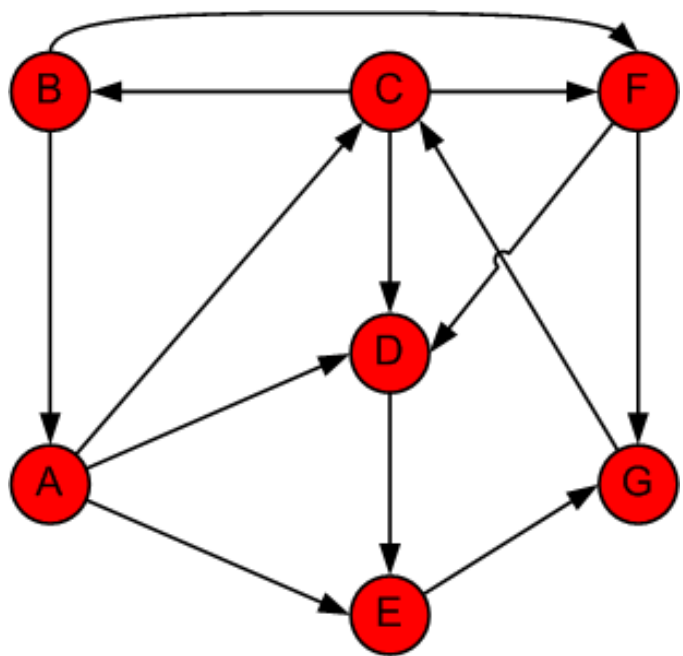


$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

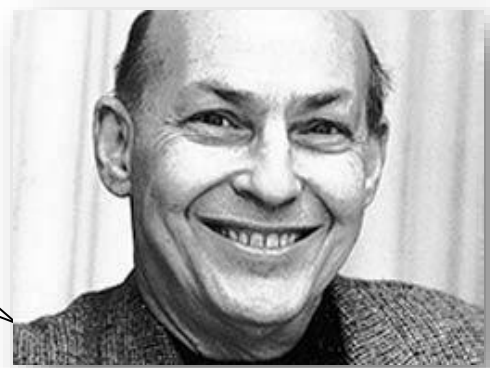
$$\mathbf{C}_p = \beta(\mathbf{I} - \alpha A^T D^{-1})^{-1} \cdot \mathbf{1} =$$

$$\begin{bmatrix} 2.14 \\ 2.13 \\ 2.14 \\ 1.45 \\ 2.13 \end{bmatrix}$$

PageRank Example – Alternative Approach [Markov Chains]



"You don't understand anything until you learn it more than one way"



Marvin Minsky (1927-2016)

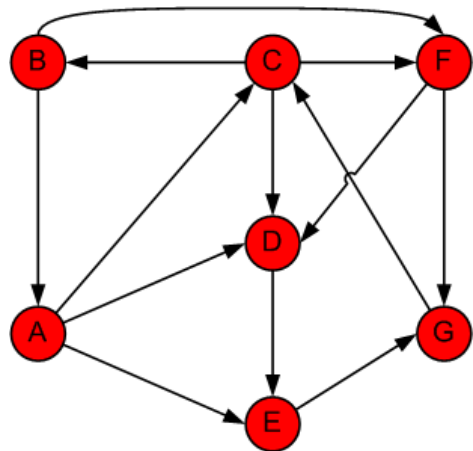
Using Power Method

$\alpha=1$ and $\beta=0$?

$$C_p(v_i) = \alpha \sum_{j=1}^n A_{j,i} \frac{C_p(v_j)}{d_j^{out}} + \beta$$

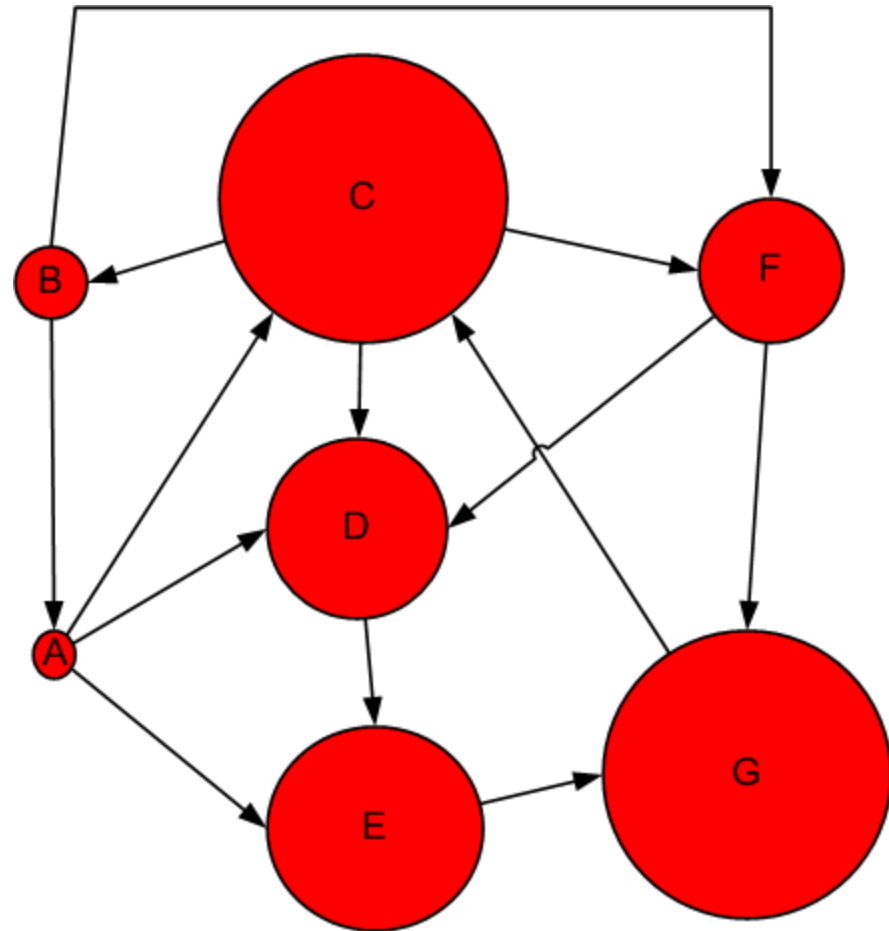
Step	A	B	C	D	E	F	G
0	1/7	1/7	1/7	1/7	1/7	1/7	1/7
1	B/2	C/3	A/3 + G	A/3 + C/3 + F/2	A/3 + D	C/3 + B/2	F/2 + E
	0.071	0.048	0.190	0.167	0.190	0.119	0.214

Effect of PageRank



PageRank

Node	Rank
A	7
B	6
C	1
D	4
E	3
F	5
G	2



Steps involved in PageRank

- Goal: generate a link of useful web pages
 - Create index of pages in response to text queries
 - Text matching with the query
 - Rank the matches
- | | |
|-------|-----------------|
| Apple | d1, d2, d8, d12 |
| Pie | d9, d3 |
| --- | -- |